Path.AI

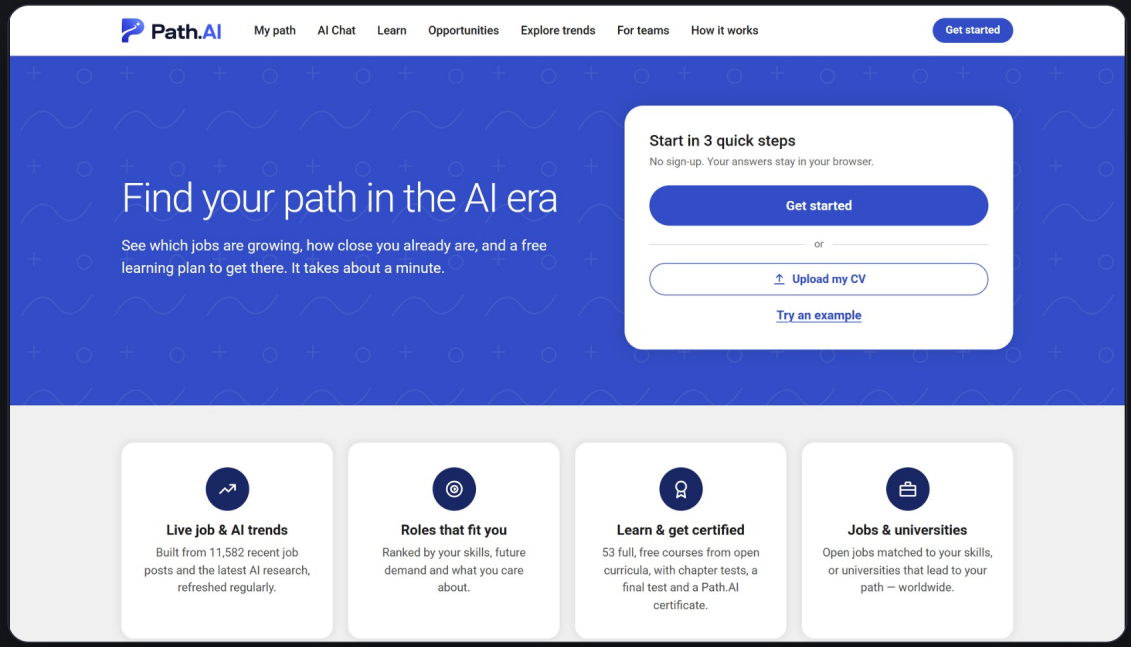
SDC XMUM HACKATHON 2026 · DOMAIN 1: THE AIVERSECITY FUTURE-READY WORKFORCE

Find your path in the AI era

Live job-market and AI-research signals become a personal next step: roles you fit, skills you're missing, a free learning plan and real openings, each explained, sourced and linked to UN SDG targets.

Team ASEA · Xiamen University Malaysia

Live prototype: pathai.lol



THE PROBLEM

AI is changing jobs faster than people can re-plan

−21%

postings for automation-prone writing and coding work, 8 months after ChatGPT

Demirci, Hannane & Zhu (2024), a global freelancing platform

600,000+

Malaysian workers need reskilling within 3–5 years

MOSTI estimate

56% vs 26%

of Malaysian professionals use AI at work vs. have been trained for it

Hays (2025)

Who Path.AI serves

Students & graduates

School leavers and fresh graduates choosing a first direction

Unemployed adults

People who need a free or low-cost route back into work

Workers facing automation

Accounts clerks, admin and customer-service staff, designers, operators

HR teams & career centres

Team skills heatmap, shareable links and printable plans

USER NEEDS & DESIGN THINKING

Designed around five questions people face

The question	What Path.AI does
“Where do I start?”	Three short steps; skills pre-filled from a job, a field of study or a CV; one tap for an example
“Which of my skills still count?”	Transferable-skill credit, a skills-match score and every gap in hours
“Can I trust this?”	Sources, dates, ranges and three AI scenarios behind every estimate
“Can I afford it in 6 hours a week?”	Free courses first, and a plan paced to the time you have each week
“Where are the jobs?”	Open postings matched to your skills, or universities in the country you choose

Our design-thinking loop

Empathise. Six fictional personas, such as Aisyah, an accounts clerk facing AI invoice processing.

Define. “As AI changes my opportunities, what should I learn next?”

Ideate. Not a generic chatbot: dated signals, rule-based scoring, AI behind guardrails.

Prototype. A live site with one-tap examples, prefilled skills and expandable reasons.

Test. 130 automated tests today; timed peer sessions with at least 20 testers in Phase 2.

THE SOLUTION

One journey from today’s job to a real opening

Skills profile → Matched roles → Skill gaps → Learning plan → Course & certificate → **Job or university**

Skills profile

Pick one of 63 jobs, a field of study, or upload a CV; every skill stays editable.

Explainable matches

Top roles by fit, demand and your values, under three AI scenarios.

Learning plan

Ordered steps paced to your weekly hours; each opens a free course.

Learn & certify

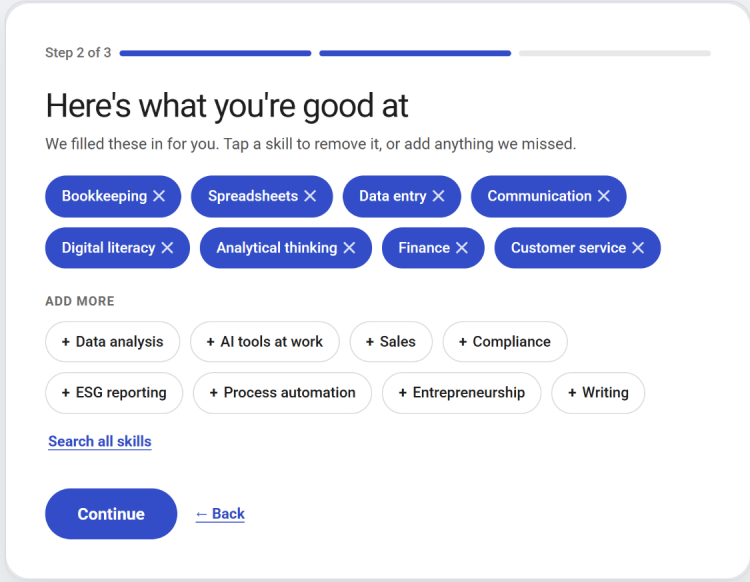
53 full courses from open curricula, chapter tests and a verifiable certificate.

Opportunities

19,481 open jobs in 94 countries, or universities in 219 countries.

AI review & AI Chat

A personal career review and a chat grounded in your own facts.



Step 2: skills arrive pre-filled from the job or CV and change with one tap.

EXAMPLE · A FICTIONAL PERSONA

Aisyah, 34: her invoices are being automated

Accounts clerk in Shah Alam, Selangor, with 9 years of bookkeeping and data entry and 6 hours a week to learn. Her company is rolling out AI invoice processing.

−16%

a year: estimated demand for her current role, Accounting / Bookkeeping Clerk

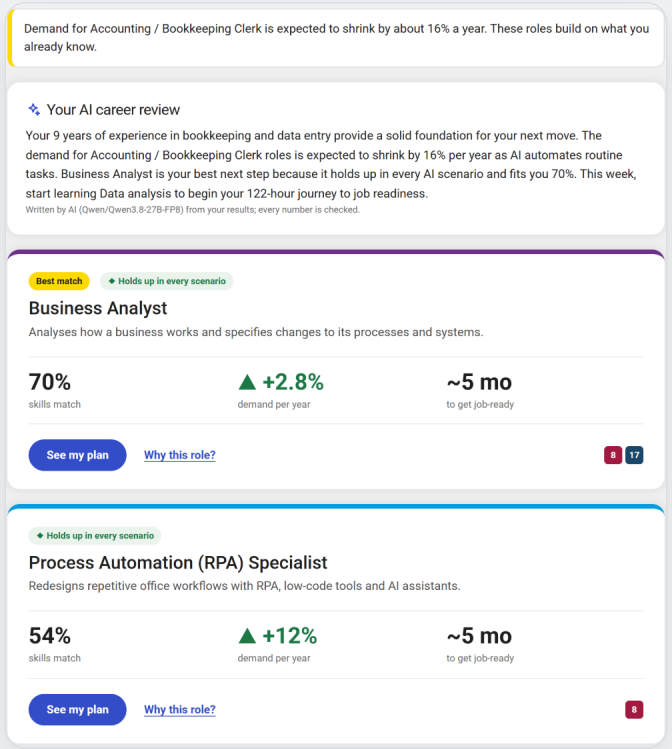
70%

skills match with Business Analyst, her best role, which holds up in every AI scenario

Fastest-growing option: Process Automation (RPA) Specialist, +12% a year. Her plan: 4 steps, about 125 hours, ~21 weeks at 6 h/week, all main courses free.

First step: AI tools & prompt engineering at work, a free 12-lesson Path.AI course with 3 chapter tests.

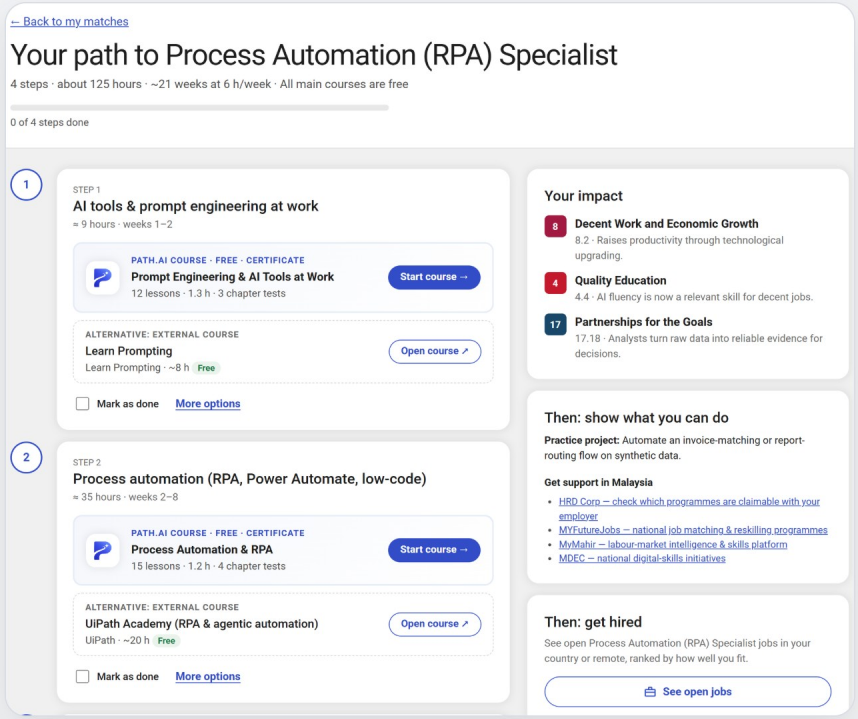
Ready to apply now: Reconciliation and Billing Specialist at Binance, 90% fit, on the employer's own page.



Live results for Aisyah, with her AI career review on top.

LEARNING PLAN & COURSES

A free course, tests and a certificate per step



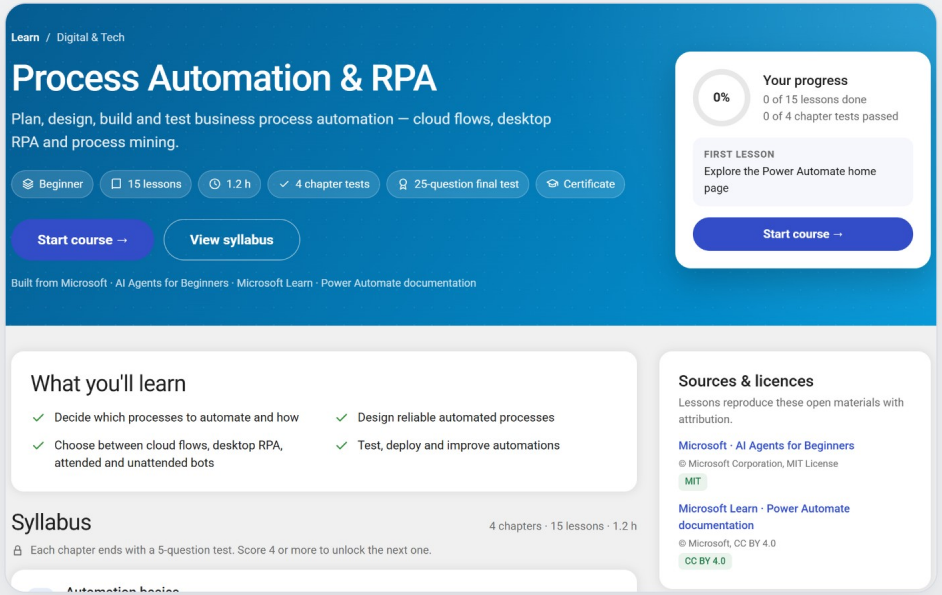
Aisyah's plan: each step opens a free Path.AI course, with an external course as the alternative.

53
free full courses

755
lessons, about 109 h

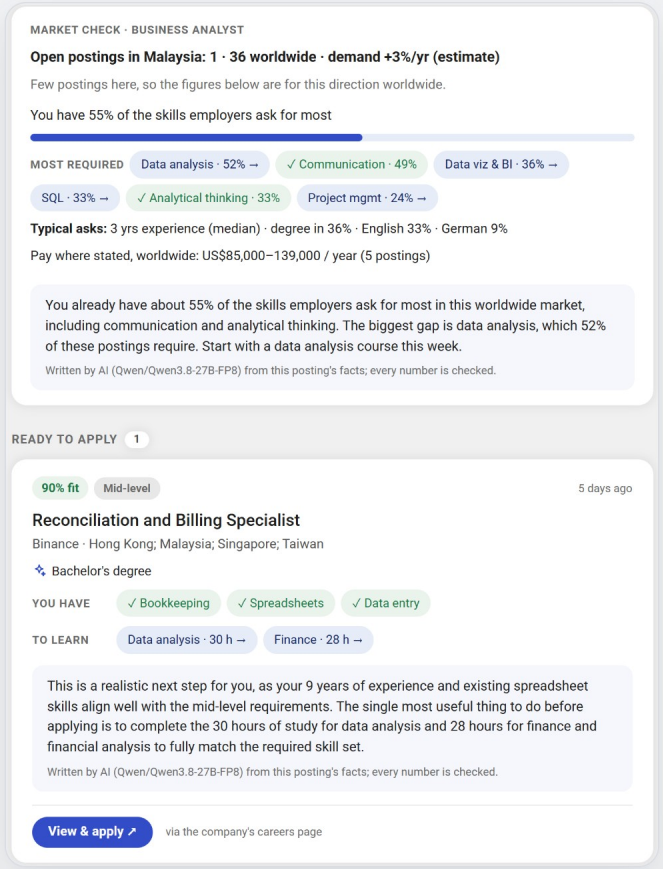
4/5
to pass each chapter test

20/25
final test, then a certificate



OPPORTUNITIES

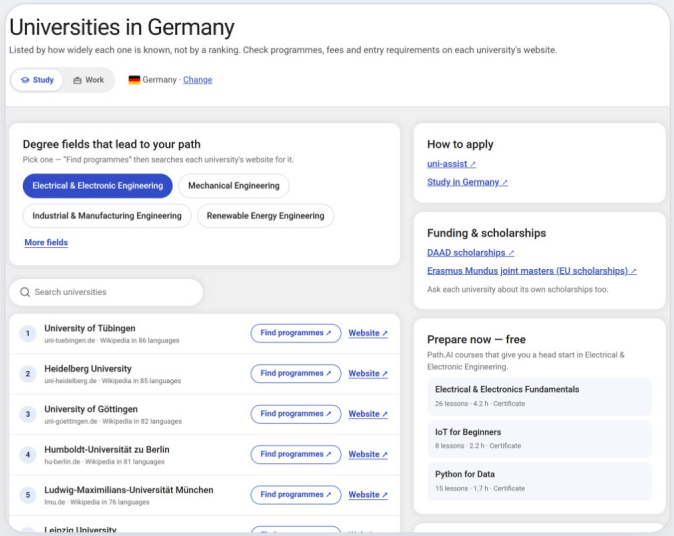
Last step: a real job or a university, anywhere



19,481
open postings in 94 countries, 6,376 remote

19,424
postings read once by AI for under $3

219
countries with university lists

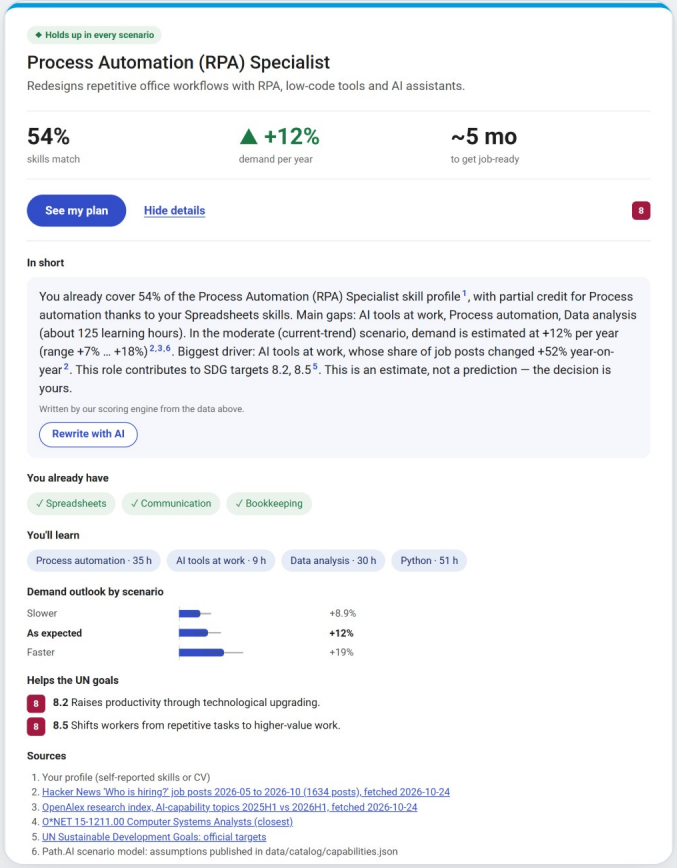


Two questions: study or work, and where?

- Jobs grouped as Ready to apply, Close or A stretch, with missing skills in hours
- Apply on the employer's own page; official portals come first
- Study: degree fields, universities and links such as UPU, PTPTN, DAAD

EXPLAINABLE RECOMMENDATIONS

Every recommendation shows its reasons



◆ Holds up in every scenario

Process Automation (RPA) Specialist

Redesigns repetitive office workflows with RPA, low-code tools and AI assistants.

54%	▲ +12%	~5 mo
skills match	demand per year	to get job-ready

See my plan | Hide details | 8

In short

You already cover 54% of the Process Automation (RPA) Specialist skill profile [1], with partial credit for Process automation thanks to your Spreadsheets skills. Main gaps: AI tools at work, Process automation, Data analysis (about 125 learning hours). In the moderate (current-trend) scenario, demand is estimated at +12% per year (range +7% … +18%) [2,3,6]. Biggest driver: AI tools at work, whose share of job posts changed +52% year-on-year [2]. This role contributes to SDG targets 8.2, 8.5 [5]. This is an estimate, not a prediction — the decision is yours.

Written by our scoring engine from the data above.

Rewrite with AI

You already have

✓ Spreadsheets | ✓ Communication | ✓ Bookkeeping

You'll learn

Process automation · 35 h | AI tools at work · 9 h | Data analysis · 30 h | Python · 51 h

Demand outlook by scenario

Slower	+8.9%
As expected	+12%
Faster	+19%

Helps the UN goals

8 **8.2** Raises productivity through technological upgrading.

8 **8.5** Shifts workers from repetitive tasks to higher-value work.

Sources

1. Your profile (self-reported skills or CV)
2. Hacker News 'Who is hiring?' job posts 2026-05 to 2026-10 (1634 posts), fetched 2026-10-24
3. OpenAlex research index, AI-capability topics 2025H1 vs 2026H1, fetched 2026-10-24
4. O*NET 15-1211.00 Computer Systems Analysts (closest)
5. UN Sustainable Development Goals: official targets
6. Path.AI scenario model: assumptions published in data/catalog/capabilities.json

Why this role?

A plain-language reason with numbered sources: your profile, dated HN job-post counts, OpenAlex research, the O*NET code, UN SDG targets and our published scenario model.

Three AI scenarios

Slower ×0.5, expected ×1 and faster ×2 AI pace, each with a range. Aisyah's RPA option grows +8.9%, +12% or +19% a year, so it is flagged as holding up in every scenario.

Live, dated data

11,582 posts from 36 monthly HN "Who is hiring?" threads (Nov 2023 – Oct 2026) and OpenAlex research on 12 AI capabilities, refreshed on 24 Oct 2026. A time machine shows any of the last 19 months.

GROUNDED AI

AI that explains and coaches, but never invents

AI career review

Four sentences from your own results: a strength, how AI affects your job, the best option and a first step.

AI Chat

Sees your profile, top roles, course progress, certificates and jobs where you look; links only to real Path.AI pages.

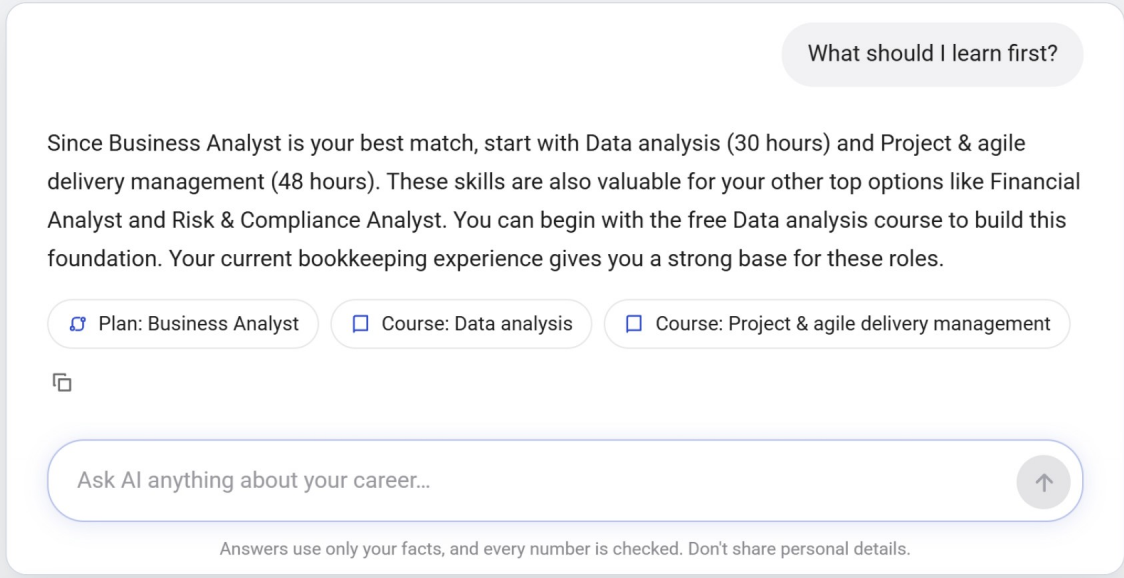
Ask AI about a job

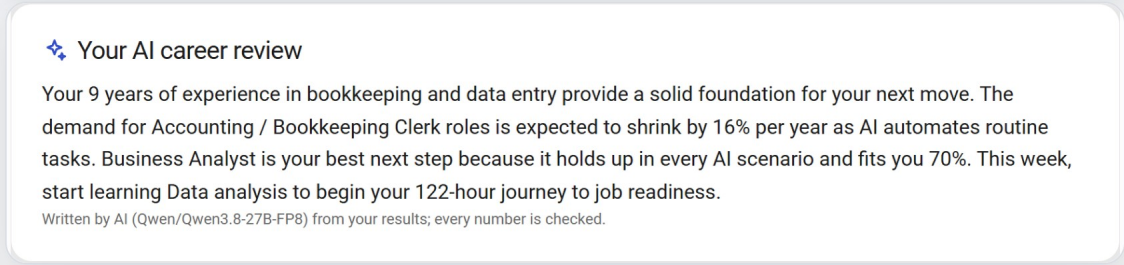
Is this posting a realistic next step, and what should you do first? Plus an AI market check for each direction.

Reads every job posting

19,424 postings read once for under $3. On 26 hand-checked ones: visa 100%, degree 96%, pay 96%, skills recall 92%.

Every number must come from Path.AI's own facts, or the answer is rejected. Each answer names the engine that wrote it.





ARCHITECTURE & ROLE OF AI

Rules compute the numbers; AI reads and explains

Live signals
HN “Who is hiring?” and OpenAlex research

Job & study data
Public job-board APIs and Wikidata

Curated catalogs
Skills, roles, courses, SDG targets

FastAPI back end · rule-based
Forecast, matching, plans, tests and job ranking: every number is computed here

Open model: Qwen3.8-27B
vLLM on RunPod, Groq as backup. Reads CVs and job posts; writes explanations and chat

Web app on pathai.lol
Vanilla JS in English, Malay and Chinese · DigitalOcean VPS in Singapore with Caddy HTTPS · profile, progress and chat stay in the browser

VALIDATION, LIMITS & PHASE 2

What we tested, what is limited, what comes next

Tested

- 130 automated tests: catalogs, engine, guardrails, Learn, jobs and API
- A fairness test: age, gender, ethnicity or location never change the ranking
- Guardrails tested against a fake model endpoint
- Every course and source link re-checked by script
- AI posting reading spot-checked on 26 postings

Limits

- HN posts are global and tech-centric, so they inform tech skills only
- Research attention is a proxy for AI progress, not adoption
- Skill sensitivities are team assumptions, still to be calibrated
- Only 166 on-site Malaysian postings in the job snapshot
- Lessons are in English; KPI targets are not yet measured

Phase 2

- Authenthication via google accounts
- Real SQL Database instead of temporary one
- More courses, universities, job opportunities.

Estimates, not predictions. AI supports; people decide.

Path.AI

Path.AI turns AI disruption into a concrete, explainable next step

Explainable

Every estimate shows its sources, dates, range and scenario

Practical

A free course for each gap, then real jobs or universities

Grounded AI

Reads and explains with checked numbers; never invents

LIVE PLATFORM

pathai.lol

CODE ON GITHUB

github.com/Almatikooo/Competition_Repo

Team ASEA · Xiamen University Malaysia · One-tap demo: pathai.lol/#/results?persona=aisyah